

Types of fractions

A proper fraction—where the numerator is smaller than the denominator—is just one type of fraction. When the number of parts is greater than the whole, the result is a fraction that can be written in two ways—either as an improper fraction (also known as a top-heavy fraction) or a mixed fraction.

$$\frac{1}{4}$$

numerator has lower value than denominator

Proper fraction

In this fraction the number of parts examined, shown on top, is less than the whole.

$$\frac{35}{4}$$

numerator has higher value than denominator

Improper fraction

The larger numerator indicates that the parts come from more than one whole.

$$10\frac{1}{3}$$

whole number

fraction

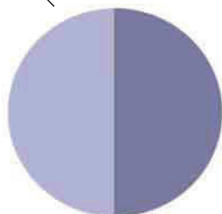
Mixed fraction

A whole number is combined with a proper fraction.

$$\frac{1}{2}$$

Half

$\frac{1}{2}$ (one half) is 1 part out of 2 equal parts in a whole.

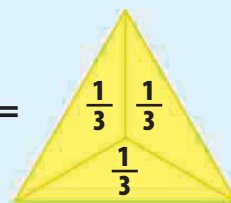


Depicting fractions

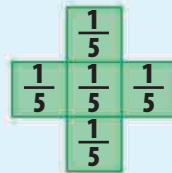
Fractions can be illustrated in many ways, using any shape that can be divided into an equal number of parts.



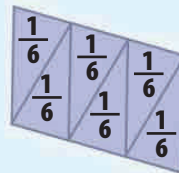
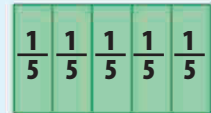
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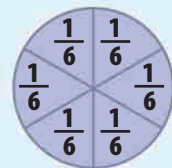
Split equally
The shapes show that there is more than one way of depicting a fraction.



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